



NAME: _____

TEACHER: _____

GOSFORD HIGH SCHOOL

2015/2016

EXTENSION 2 MATHEMATICS

HSC ASSESSMENT TASK 1.

Time Allowed: 60 minutes (plus 5 min. reading time)

- Write using black or blue pen.
- Board-approved calculators may be used.
- Section III should be started on a new page and Section IV should be started on a new page.
- All necessary working should be shown in Section II, III and IV.

SECTION	QUESTION TYPE	MARKS	RESULT
I	MULTIPLE CHOICE	4	
II	EXTENDED RESPONSE	18	
III	EXTENDED RESPONSE	18	
	TOTAL	40	

SECTION I. (4 marks) Answer on the multiple choice answer sheet.

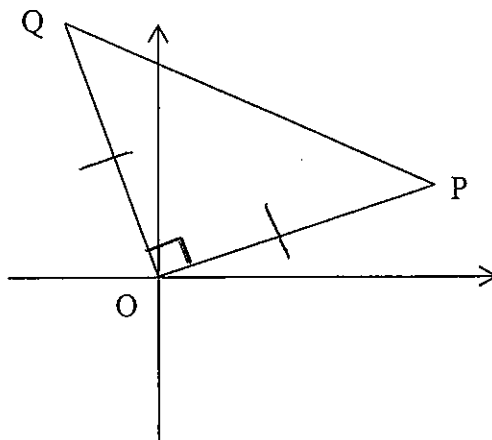
1. If $w = -1 - \sqrt{3}i$ what is the value of $\arg(w)$?

- A. $\frac{\pi}{3}$ B. $\frac{-\pi}{3}$ C. $\frac{2\pi}{3}$ D. $\frac{-2\pi}{3}$

2. On an Argand diagram, the points A and B represent the complex numbers $3 + 2i$ and $2 - i$ respectively. The point P is such that $OAPB$ is a parallelogram. What complex number does P represent?

- A. $5 + i$ B. $5 - i$ C. $-1 - 3i$ D. $1 + 3i$

3. The points P and Q in the complex plane correspond to the complex numbers z and w respectively. The triangle OPQ is isosceles with $OP = OQ$ and $\angle POQ$ is right angle.



Which of these statements is correct?

- A. $z^2 + w^2 = 1$. B. $z^2 - w^2 = 1$. C. $w^2 - z^2 = 1$. D. $z^2 + w^2 = 0$.

4. The cartesian equation of the locus specified by $|z|^2 = z + \bar{z}$ is

- A. $(x - 1)^2 + y^2 = 1$. B. $(x + 1)^2 + y^2 = 1$
C. $x^2 + (y - 1)^2 = 1$ D. $x^2 + (y + 1)^2 = 1$

SECTION II. (18 marks)

1. If $z = 2 + i$ and $\omega = 1 - 2i$. Find in the form $x + iy$, where x and y are real,

(i) $2z + i\omega$ (1)

(ii) $z\bar{\omega}$ (1)

(iii) $\frac{z}{\omega}$ (2)

2. (i) Express $1 - \sqrt{3}i$ in modulus-argument form. (2)

(ii) Express $(1 - \sqrt{3}i)^5$ in modulus-argument form. (1)

(iii) Hence, express $(1 - \sqrt{3}i)^5$ in the form $x + iy$, where x and y are real. (2)

3. On separate Argand diagrams sketch each of the following regions.

(i) $\operatorname{Re}(z) \geq -1$ and $1 < \operatorname{Im}(z) < 2$. (1)

(ii) $|z + 2i| \leq 2$. (1)

(iii) $z + \bar{z} > 6$, $|z - 1| \leq 4$ and $|\arg z| < \frac{\pi}{6}$. (2)

4. Given $z = r(\cos\theta + i\sin\theta)$, where $z \neq 0$:

(i) Show that $z\bar{z}$ is real. (1)

(ii) Use De Moivre's Theorem to show that $z^n + \bar{z}^n$ is real for all integers $n \geq 1$. (1)

(iii) Show that $\frac{z}{\bar{z}} + \frac{\bar{z}}{z}$ is real. (2)

(iv) Hence, show that $-2 \leq \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \leq 2$. (1)

SECTION III. (18 marks) Start a new page.

1. (i) Find all pairs of integers x and y such that $(x + iy)^2 = -3 - 4i$. (2)

(ii) Hence, solve $z^2 - 3z + (3 + i) = 0$, expressing the roots in the form $a + ib$, where a and b are real. (2)

2. (i) Solve the equation $z^3 - 1 = 0$ expressing your answers in modulus-argument form. (1)

(ii) Let w be the root of $z^3 - 1 = 0$ with the smallest positive argument.

Simplify $(1 + w^2)^4$. (2)

3. Find the three cube roots of $2 - 2i$. (4)

4. Sketch the locus of z and find the cartesian equation of the locus if:

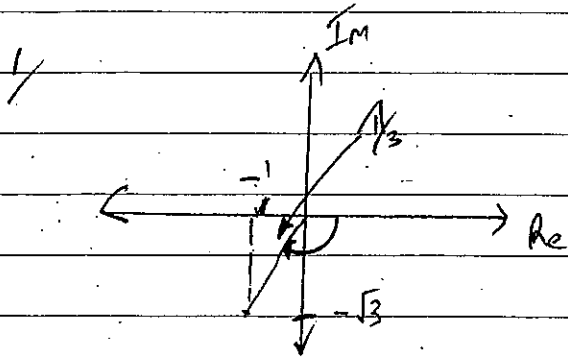
$$|z - 3 - 2i| = |z - 5 + 3i|. \quad (3)$$

5. If $z = \cos\theta + i \sin\theta$ and using the expansion of $(z - \frac{1}{z})^4$, show that

$$\sin^4\theta = \frac{1}{8} (\cos 4\theta - 4 \cos 2\theta + 3). \quad (4)$$

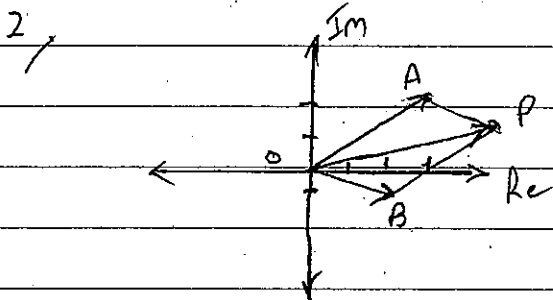
SOLUTIONS

SECTION 1.



$$\arg(w) = -\frac{2\pi}{3}$$

(D)



$$\begin{aligned}\vec{OP} &= \vec{OA} + \vec{OB} \\ &= 3+2i + 2-i \\ &= 5+i\end{aligned}$$

$$P \text{ is } 5+i$$

(A)

3/

$$\begin{aligned}w &= iz \\ w^2 &= i^2 z^2 \\ &= -z^2 \\ \therefore z^2 + w^2 &= 0\end{aligned}$$

(D)

4/

$$|z| = \sqrt{x^2 + y^2}$$

$$\begin{aligned}z + \bar{z} &= x+iy + x-iy \\ &= 2x\end{aligned}$$

1/

$$|z|^2 = z + \bar{z}$$

$$x^2 + y^2 = 2x$$

$$x^2 - 2x + y^2 = 0$$

$$x^2 - 2x + 1 + y^2 = 1$$

$$(x-2)^2 + y^2 = 1$$

(A)

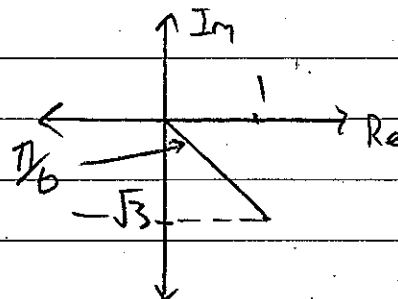
SECTION II

$$\begin{aligned} 1/(i) \quad 2z + i\omega &= 2(2+i) + i(1-2i) \\ &= 4 + 2i + i - 2i^2 \\ &= 6 + 3i \end{aligned} \quad (1)$$

$$\begin{aligned} (ii) \quad z\bar{\omega} &= (2+i)(1+2i) \\ &= 2 + 4i + i + 2i^2 \\ &= 0 + 5i \end{aligned} \quad (1)$$

$$\begin{aligned} (iii) \quad \frac{z}{\omega} &= \frac{(2+i)}{(1-2i)} \times \frac{(1+2i)}{(1+2i)} \\ &= \frac{0+5i}{1-4i^2} \\ &= \frac{0+5i}{5} \\ &= 0+i \end{aligned} \quad (2)$$

2/ (i)

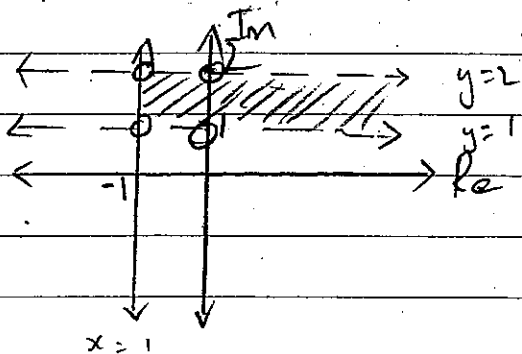

$$\begin{aligned} |1-\sqrt{3}i| &= \sqrt{1^2 + \sqrt{3}^2} \\ &= 2 \\ \arg(1-\sqrt{3}i) &= -\frac{\pi}{3} \end{aligned}$$

$$\therefore 1-\sqrt{3}i = 2 \operatorname{cis} \left(-\frac{\pi}{3}\right) \quad (2)$$

$$\begin{aligned} (ii) \quad (1-\sqrt{3}i)^5 &= 2^5 \operatorname{cis} \left(-\frac{5\pi}{3}\right) \\ &= 32 \operatorname{cis} \frac{\pi}{3} \end{aligned} \quad (1)$$

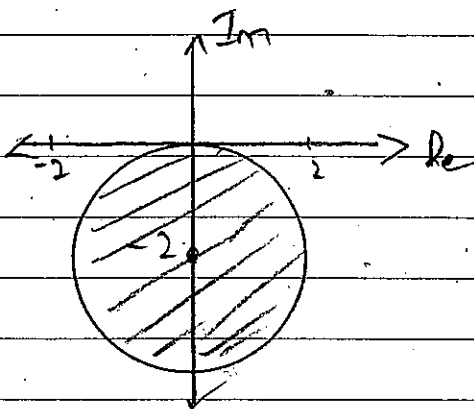
$$\begin{aligned} (iii) \quad (1-\sqrt{3}i)^5 &= 32 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\ &= 32 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \\ &= 16 + 16\sqrt{3}i \end{aligned} \quad (2)$$

3/ (i) If $\operatorname{Re}(z) \geq -1$, $x \geq -1$
 If $1 < \operatorname{Im}(z) < 2$, $1 < y < 2$



(1)

(ii) $|z + 2i| \leq 2$
 $|z - (0 - 2i)| \leq 2$

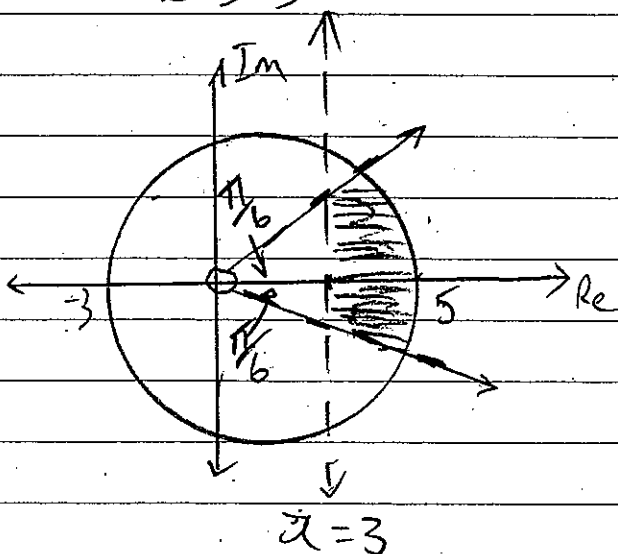


(1)

(iii) $z + \bar{z} > 6$
 $2x > 6$
 $x > 3$

$|z - 1| \leq 4$
 $|z - (1 + 0i)| \leq 4$

$-\frac{\pi}{6} < \arg(z) < \frac{\pi}{6}$



(2)

$$4/ \quad (i) \quad Z = r(\cos \theta + i \sin \theta) \\ \bar{Z} = r(\cos \theta - i \sin \theta)$$

$$\begin{aligned} Z\bar{Z} &= r(\cos \theta + i \sin \theta) \times r(\cos \theta - i \sin \theta) \\ &= r^2 (\cos^2 \theta - i^2 \sin^2 \theta) \\ &= r^2 (\cos^2 \theta + \sin^2 \theta) \\ &= r^2 \end{aligned} \quad (1)$$

which is real.

$$(ii) \quad Z^n = r^n (\cos n\theta + i \sin n\theta)$$

$$\bar{Z}^n = r^n (\cos n\theta - i \sin n\theta)$$

$$\begin{aligned} Z^n + \bar{Z}^n &= r^n (2 \cos n\theta) \\ &= 2r^n \cos n\theta \end{aligned} \quad (1)$$

which is real.

$$(iii) \quad \frac{Z}{Z} + \frac{\bar{Z}}{Z}$$

$$= \frac{Z^2 + \bar{Z}^2}{Z\bar{Z}}$$

$$= \frac{2r^2 \cos 2\theta}{r^2} \quad \begin{array}{l} \text{from (ii)} \\ \text{from (i)} \end{array}$$

$$= 2 \cos 2\theta$$

which is real.

$$(iv) \quad \text{Since } -1 \leq \cos 2\theta \leq 1 \quad (1) \\ -2 \leq 2 \cos 2\theta \leq 2$$

$$\therefore -2 \leq \frac{Z}{Z} + \frac{\bar{Z}}{Z} \leq 2$$

SECTION III

$$\angle (i) \quad \text{If } (x+iy)^2 = -3-4i$$
$$x^2 - y^2 + 2xyi = -3-4i$$

$$\therefore x^2 - y^2 = -3$$

$$\& \quad 2xy = -4$$

$$y = -\frac{2}{x}$$

$$\therefore x^2 - \left(-\frac{2}{x}\right)^2 = -3$$

$$x^2 - \frac{4}{x^2} = -3$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2 + 4)(x^2 - 1) = 0$$

$$\therefore x^2 = 1$$

$$x = \pm 1$$

$$y = \mp 2$$

(2)

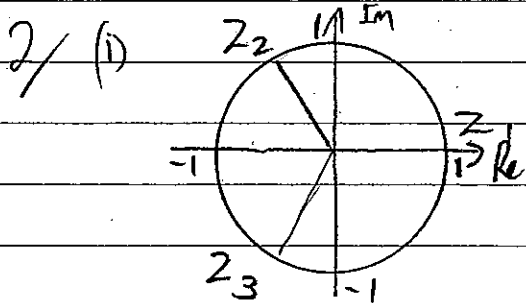
$$(ii) \quad \text{If } z^2 - 3z + (3+i) = 0$$
$$z = \frac{3 \pm \sqrt{9 - 4(3+i)}}{2}$$

$$= \frac{3 \pm \sqrt{-3-4i}}{2}$$

$$= \frac{3 \pm (1-2i)}{2}$$

(2)

$$= 2-i \quad \text{or} \quad 1+i$$



$$\text{If } z^3 = 1$$

$$z = 1, \quad \text{cis } \frac{2\pi}{3}, \quad \text{cis } -\frac{2\pi}{3}$$

(1)

(ii) Let the roots be $1, \omega, \omega^2$

$$\sum \text{roots} = -\frac{b}{a}$$

$$\therefore 1 + \omega + \omega^2 = 0$$

$$1 + \omega^2 = -\omega$$

$$\begin{aligned} \text{So } (1 + \omega^2)^4 &= (-\omega)^4 & (2) \\ &= \omega^4 \\ &= \omega^3 \times \omega \\ &= 1 \times \omega \\ &= \omega \end{aligned}$$

3. If $z^3 = 2 - 2i$
 $|z|^3 = \sqrt{2^2 + (-2)^2}$
 $= 2\sqrt{2}$
 $\therefore |z| = \sqrt{2}$

Let the roots be of the form $\sqrt{2} \operatorname{cis} \phi$

$$\text{So } z^3 = (\sqrt{2} \operatorname{cis} \phi)^3$$

$$z^3 = 2\sqrt{2} \operatorname{cis} 3\phi$$

$$\therefore 2\sqrt{2} (\cos 3\phi + i \sin 3\phi) = 2 - 2i$$

Equate real & imaginary parts

$$2\sqrt{2} \cos 3\phi = 2$$

$$\cos 3\phi = \frac{1}{\sqrt{2}}$$

$$2\sqrt{2} \sin 3\phi = -2$$

$$\sin 3\phi = -\frac{1}{\sqrt{2}}$$

$$\therefore 3\phi = \frac{7\pi}{4}, \frac{15\pi}{4}, \frac{23\pi}{4}$$

$$\phi = \frac{7\pi}{12}, \frac{15\pi}{12}, \frac{23\pi}{12}$$

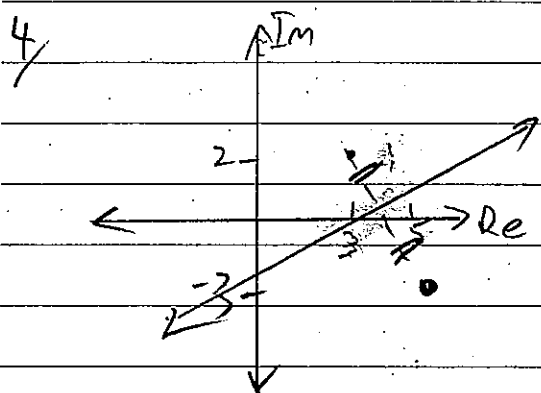
$$\begin{array}{c|c} S & A \\ \hline T & C \end{array} \checkmark$$

∴ the 3 cube roots are

(4)

$$z = \sqrt{2} \cos \frac{7\pi}{12}, \quad \sqrt{2} \cos \frac{-9\pi}{12}, \quad \sqrt{2} \cos \frac{-\pi}{12}$$

$$\text{i.e. } z = \sqrt{2} \cos \frac{\pi}{12}, \quad \sqrt{2} \cos \frac{-3\pi}{4}, \quad \sqrt{2} \cos \frac{-\pi}{12}$$



$$\text{If } |z - 3 - 2i| = |z - 5 + 3i|$$

$$|z - (3 + 2i)| = |z - (5 - 3i)|$$

The locus is the perp. bisector of the interval joining $(3, 2)$ & $(5, -3)$ on the complex plane

Let $z = x + iy$

$$|x + iy - 3 - 2i| = |x + iy - 5 + 3i|$$

$$\sqrt{(x-3)^2 + (y-2)^2} = \sqrt{(x-5)^2 + (y+3)^2}$$

(3)

$$\cancel{x^2} - 6x + 9 + y^2 - 4y + 4 = \cancel{x^2} - 10x + 25 + \cancel{y^2} + 6y + 9$$

$$\therefore 4x - 10y - 21 = 0.$$

5/

$$\left(z - \frac{1}{z}\right)^4 = z^4 - 4z^3 \cdot \frac{1}{z} + 6z^2 \cdot \frac{1}{z^2} - 4z \cdot \frac{1}{z^3} + \frac{1}{z^4}$$

$$= \left(\frac{z^4 + 1}{z^4}\right) - 4\left(\frac{z^2 + 1}{z^2}\right) + 6$$

Since $z^n + \frac{1}{z^n} = 2 \cos n\theta$

$$\left(z - \frac{1}{z}\right)^4 = 2 \cos 4\theta - 8 \cos 2\theta + 6$$

$$\text{Also } z^n - \frac{1}{z^n} = 2i \sin n\theta$$

$$\begin{aligned}\therefore \left(2 - \frac{1}{2}\right)^4 &= (2i \sin \theta)^4 \\ &= 16 i^4 \sin^4 \theta \\ &= 16 \sin^4 \theta\end{aligned}$$

4

$$\therefore 16 \sin^4 \theta = 2 \cos 4\theta - 8 \cos 2\theta + 6$$

$$\begin{aligned}\sin^4 \theta &= \frac{1}{8} \cos 4\theta - \frac{1}{2} \cos 2\theta + \frac{3}{8} \\ &= \frac{1}{8} [\cos 4\theta - 4 \cos 2\theta + 3]\end{aligned}$$